

Independent Audit of Aegis Finance

The bottom line is that the current evidence does **not** yet show that the surviving strategies are real rather than the expected output of a 179-trial search under weak multiple-testing control, and the documented leaks mean even the “confirm” sample cannot be treated as clean until the whole data lineage is re-audited. The good news is that the project already has more discipline than most retail quant efforts; the bad news is that several of its strongest process claims appear softer than machine-enforced, and that is exactly where false discoveries survive (Bailey & De Prado, 2014; Hansen, 2005; Clarke et al., 2020).

Validation

VERIFIED. The claimed “noise best-of-179 Sharpe ≈ 0.71 on 15 years” is directionally plausible, not obviously absurd. Under the Bailey–López de Prado framing, selection bias rises with the number of trials and can make even impressive backtests look false unless sample length and non-normality are accounted for (Bailey & De Prado, 2014; Bailey & De Prado, 2021). Harvey, Liu, and Zhu argue that in a crowded factor-search environment the classical $t > 2$ hurdle is too low, and that new factors need t-statistics above 3.0 to clear a modern multiple-testing bar (Harvey, 2014; Harvey et al., 2016). Bailey and López de Prado also make the stronger point that, with enough trials, no fixed Sharpe threshold alone can rule out a false strategy (Bailey & De Prado, 2021).

A compact check is to translate annualized Sharpe to a t-stat over 15 yearly-equivalents: ($t \approx SR \sqrt{T}$). If $SR = 0.71$ and $T = 15$, $t \approx 2.75$, which is just above the project’s stated $Z(179)=2.729$ threshold and still below the more general “new factor >3.0 ” hurdle from Harvey-Liu-Zhu. **So the arithmetic is internally coherent, but the interpretation is too lenient:** clearing 2.73 in one screened family is weaker than clearing the broader asset-pricing hurdle, and both are before leak correction. That is especially important because as few as three independent trials can already make an annualized Sharpe of 1 statistically unconvincing in DSR terms (Bailey & De Prado, 2021).

Evidence-Strength vs Claim for the main validation questions:

Evidence Strength

Claim



Multiple testing materially inflates backtest performance, and DSR/related corrections are necessary after broad search (Bailey & De Prado, 2014; Hansen, 2005)



Walk-forward alone is weak protection against overfitting relative to CPCV-style methods in synthetic finance settings (Arian et al., 2024)



TSMOM-XA deserves skepticism because pooled evidence for classic TSM is weaker than the headline literature suggests (Huang et al., 2020)

FIGURE 1 Evidence strength for core validation claims

VERIFIED prior: before seeing the dossier’s placebo dispute, the expectation was that pooled t-statistics across many random seeds would overstate certainty if seed outputs share the same underlying history. That prior fits the literature: dependence-aware procedures outperform naive ones, and Romano–Wolf stepdown is more powerful than Bonferroni precisely because it respects dependence among test statistics (Clarke et al., 2020; Romano & Wolf, 2016). So on the placebo gate, **seed-level t is the safer default than pooled t** if seeds are just Monte Carlo perturbations of execution, hyperparameters, or bootstrap resamples on the same economic path. A third reading is better still: treat the placebo family as a multiple-testing problem and test whether the real strategy is superior to the placebo universe with Hansen’s SPA, not whether one summary t-stat beats zero (Hansen, 2005).

The right discriminator between “real survivors” and “what luck after 179 trials looks like” is a **locked resampling context**. Freeze three survivors exactly as specified, generate a matched library of 179 placebo or structurally-related alternatives, and run CPCV or CSCV with purging and embargo so each fold replays the original selection problem (Arian et al., 2024; Fonseca, 2026). Then apply SPA to test whether the best survivor is genuinely better than the benchmark family, and Romano–Wolf across the three finalists to control familywise error on “gp-small,” “fusion,” and “TSMOM-XA” jointly (Hansen, 2005; Clarke et al., 2020).

Methodology Review

The explore/confirm split is a real strength because chronological separation is better than recycling the same full sample, and pre-registration of gates is worth keeping (Pham et al., 2026). But a top referee would still reject the current version if “confirm” reused any tainted feature store, any silently revised linkage, or any post-hoc audit-discovered bug fix, because point-in-time correctness is a filtration property of the whole system, not a property of the final notebook (Fonseca, 2026).

The project is also underusing the distinction between **described discipline** and **enforced discipline**. The literature on open-source cross-sectional replication shows that tiny implementation choices and coding errors can materially move t-stats, even when the broad result survives (Chen & Zimmermann, 2021). That makes the registry defect—18 recorded trials versus 179 actual trials—not just a counter bug but a proof that the audit trail is not yet the authoritative source of truth.

The likely over-disciplined part is the placebo gate if it uses **random-date placebos divorced from the economic null**. Hansen explicitly notes that large universes of poor or irrelevant alternatives can reduce power, and SPA was designed to be less sensitive to exactly that problem (Hansen, 2005). A random-date placebo can kill true signals if it generates nulls that are easier to beat than the real deployment problem or, conversely, if dependence is mishandled and the placebo family is effectively counted too many times.

Minimum protocol changes with maximum payoff:

Change	Exact Decision It Changes	Why It Matters
Replace WFA-first gating with CPCV/CSCV first	Promotion from explore to confirm	WFA is weaker against false discoveries than CPCV in synthetic comparisons (Arian et al., 2024)
Add SPA + Romano–Wolf	“Does any survivor beat the screened family?”	Converts ad hoc significance into dependence-aware family testing (Hansen, 2005; Clarke et al., 2020)

Change	Exact Decision It Changes	Why It Matters
Enforce dataset-as-of manifests	Every feature build and rerun	Makes PIT auditable and catches FRED/BoardEx-style time leaks (Fonseca, 2026)
Separate research registry from code execution logs	Trial counts and prereg status	Prevents “described but not executed” discipline failures (Pham et al., 2026)

FIGURE 2 Highest-payoff protocol changes for the pipeline

Source Additions

The single most relevant public codebase to mine harder is **Chen-Zimmermann’s open-source cross-sectional asset pricing stack**. Take the characteristic definitions and replication harness, but more importantly take its norm that every signal is a reproducible transformation with explicit implementation choices, because their reproductions recover 98% of clearly significant predictors and expose how often disagreement is really about definitions, weighting, and breakpoints (Chen & Zimmermann, 2021).

For methodology, add **Hansen SPA**, **Romano–Wolf stepdown**, and **CPCV/CSCV** immediately. SPA answers “is the chosen rule genuinely superior after searching many rules?” (Hansen, 2005). Romano–Wolf answers “which of several survivors still stand after dependence-aware multiple testing?” (Clarke et al., 2020). CPCV/CSCV answers “how fragile is the ranking to resampled chronology?” and has lower PBO than walk-forward in modern tests (Arian et al., 2024; Fonseca, 2026).

For PIT and LLM leakage, the underused papers here are the 2026 look-ahead benchmarks. Standard LLMs show large in-sample gains that collapse out-of-sample when memorization is removed, while point-in-time models generalize better (Benhenda, 2026). FinCAD-style mitigation cuts in-sample LLM backtest returns sharply on memorized dates while leaving truly out-of-sample performance roughly unchanged (Li et al., 2026). Any LLM news/social feature should therefore be forced through a **numeric extraction path plus anti-memorization ablation** before it enters the signal library.

High-value datasets you may already be paying for but are probably underusing are **IBES analyst coverage/history** and **BoardEx full network history**. Fang and Peress show neglected stocks with no media coverage earn higher returns, especially among small, low-analyst-followed, high-idiosyncratic-volatility firms (Fang & Peress, 2008). BoardEx supports point-in-time director network measures, and director connectedness has been linked to more profitable insider trades (Goergen & Renneboog, 2018).

Thesis Review

The owner’s **social-bias thesis** has real backing, but most of the strongest evidence lives in neglected, hard-to-arbitrage names and often in the short leg. Low media coverage predicts higher subsequent returns (Fang & Peress, 2008). Underreaction-related anomalies are much stronger among low-media-coverage firms, with average five-factor alpha spreads around 0.97% per month versus 0.24% for high-coverage firms, but most of that comes from the short leg and harder-to-arbitrage stocks (Chen et al., 2022). Sin stocks fit the same neglect channel: they are less institutionally held, less followed by analysts, and outperform after controls (Hong & Kacperczyk, 2009).

That same evidence is also the attack on the thesis. If your deployment is long-only retail, and \$28 already showed one spread lived almost entirely in the short leg, then many attention and neglect anomalies will shrink dramatically in tradable form (Chen et al., 2022). Gross profitability is stronger as a long-only candidate than “attention shorts”: Novy-Marx finds gross profits-to-assets has power comparable to book-to-market, and controlling for profitability especially improves value among large, liquid stocks (Novy-Marx, 2013). But even there, profitability anomalies have attenuated, and some of that attenuation appears tied to changes in fundamentals rather than arbitrage alone (Choy et al., 2023).

For politician and insider flows, the honest synthesis is **mixed and regime-specific**. Early Senate and House studies found large abnormal returns from mimicking purchases (Ziobrowski et al., 2004; Ziobrowski et al., 2011). Post-STOCK-Act evidence is much weaker in aggregate, finding no broad superior performance from 2012–2020 (Belmont et al., 2022). A more recent nuance is that returns appear after **disclosure**, not necessarily after the actual trade, implying a replicate-the-disclosure effect rather than pure private information extraction (Lazzaletto, 2024). Politically connected insiders remain more compelling than politicians as a class: their trades were more informative around TARP and predicted announcement reactions up to 30 days in advance (Jagolinzer et al., 2020).

Ideas worth extending first:

Idea	Data And PIT Rule	Null And Control	Capacity
BoardEx contagion graph for insider clusters	BoardEx role start/end dates only; no `Curr` sentinel	Null: no return drift after clustered buys by connected directors; control with shuffled network edges	Moderate in mid/small caps (Goergen & Renneboog, 2018)
Disclosure-following politician signal	STOCK Act filing timestamp, not trade date	Null: no abnormal return after disclosure; placebo with matched non-political disclosures	Low–moderate (Lazzaletto, 2024)
Coverage-gap profitability long-only	IBES analyst history + media counts + gross profitability	Null: profitability works no better in neglected names; placebo with future analyst counts forbidden	Moderate retail capacity (Novy-Marx, 2013; Fang & Peress, 2008)
LLM attention extractor with honesty-drop test	Dated articles only; out-of-sample beyond model cutoff	Null: feature disappears after anti-memorization penalty	Unknown until tested (Benhenda, 2026; Li et al., 2026)

FIGURE 3 Specific new idea designs with controls

Roadmap

The 37-day paper lane is **not** meaningful evidence. That framing is correct. Short-horizon Sharpe estimates are extremely noisy, and the literature’s broader lesson is that even decade-long backtests can be false after selection, so 37 trading days should be treated only as an operational smoke test (Bailey & De Prado, 2021). A reasonable minimum before any directional claim is **12 months for operational robustness** and closer to **24 months** for any skill claim, which is consistent with the project’s own conservatism and with the instability of short-sample Sharpe ratios implied throughout the overfitting literature (Falck et al., 2021).

The 90-day plan would be:

1. **Freeze the entire current stack.** Dependency: none. Kill criterion: if exact rerun of every survivor is not byte-identical, stop all research.
2. **Build as-of manifests for every dataset and feature table.** Dependency: frozen stack. Kill criterion: if any confirm-period feature cannot prove publication timestamp, invalidate that trial family.
3. **Re-run 179-trial family with CPCV/CSCV + SPA + Romano–Wolf.** Dependency: manifests. Kill criterion: if no survivor survives familywise control, kill all survivors as deployable claims.
4. **Audit TSMOM-XA first.** Prior: weakest survivor because classic TSM evidence is mixed and pooled regressions overstate reliability (Huang et al., 2020). Kill criterion: if advantage over historical-mean or vol-targeted baseline disappears, retire it.
5. **Refactor gp-small into long-only neglected-profitability sleeves.** Dependency: clean PIT fundamentals plus analyst/media data. Kill criterion: if alpha lives mostly in the short leg or microcaps, do not fund it.
6. **Ban unrestricted LLM signals until honesty-drop testing exists.** Dependency: none. Kill criterion: if anti-memorization penalties erase in-sample edge, classify as leak-driven.
7. **Prototype only two new idea families:** BoardEx insider clusters and disclosure-following politician trades. Dependency: PIT event timestamps. Kill criterion: if either fails a preregistered placebo and costs screen, archive it in the negative ledger.
8. **Size live capital by fractional Kelly on conservative expected Sharpe,** never by backtest peak Sharpe. Kelly maximizes long-run growth but full Kelly is too fragile; fractional Kelly is the right aggressive-but-not-innumerate compromise (MacLean et al., 2011; Brockwell, 2021).

The single most important shift is from **strategy-level significance** to **pipeline-level falsification**: the project should stop asking whether a survivor backtest looks good and start asking whether the full searched-and-tainted research process can still produce that survivor after PIT, dependence-aware testing, and anti-leak audits. The single biggest open question is whether **gp-small** retains a long-only, post-cost edge once the short leg, microcap dependence, and confirm-period data lineage are all cleaned.

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